Python Lists

Python Lists

- A list is a collection which is ordered and changeable. In Python lists are written with square brackets.

Example:

```
thislist = ["apple", "banana", "cherry"]
print(thislist)
```

Output:

```
['apple', 'banana', 'cherry']
```

Python Lists

- Example: Print the second item of the list:

```
thislist = ["apple", "banana", "cherry"]
print(thislist[1])
```

Output:

```
banana
```

- Negative indexing means beginning from the end, -1 refers to the last item, -2 refers to the second last item etc.

```
thislist = ["apple", "banana", "cherry"]
print(thislist[-1])   # output: cherry
```

Python Lists

- You can specify a range of indexes by specifying where to start and where to end the range.
- When specifying a range, the return value will be a new list with the specified items.

Example :

```
thislist = ["apple", "banana", "cherry", "orange", "kiwi", "melon",
    "mango"]
print(thislist[2:5])
```

Output : ['cherry', 'orange', 'kiwi']

The search will start at index 2 (included) and end at index 5 (not included).

Python Lists

Example :This example returns the items from the beginning to "orange“

```
thislist = ["apple", "banana", "cherry", "orange", "kiwi", "melon",
  "mango"]
  print(thislist[:4])
```

#output : ['apple', 'banana', 'cherry', 'orange']

Example :This example returns the items from "cherry" and to the end:

```
thislist = ["apple", "banana", "cherry", "orange", "kiwi", "melon",
  "mango"]
  print(thislist[2:])
```

#output : ['cherry', 'orange', 'kiwi', 'melon', 'mango']

Python Lists

- Specify negative indexes if you want to start the search from the end of the list:

Example : This example returns the items from index -4 (included) to index -1 (excluded)

```
thislist = ["apple", "banana", "cherry", "orange", "kiwi", "melon",
    "mango"]
     print(thislist[-4:-1])

#output : ['orange', 'kiwi', 'melon']
```

Python Lists

- To change the value of a specific item, refer to the index number:

Example : Change the second item:

```
thislist = ["apple", "banana", "cherry"]
thislist[1] = "blackcurrant"
print(thislist)
```

```
#output : ['apple', 'blackcurrant', 'cherry']
```

Python Lists

- Print all items in the list, one by one:

Example :

```
thislist = ["apple", "banana", "cherry"]
for x in thislist:
    print(x)
```

```
#output : apple
          banana
          cherry
```

Python Lists

- Check if "apple" is present in the list::

Example :

```
thislist = ["apple", "banana", "cherry"]
if "apple" in thislist:
    print("Yes, 'apple' is in the fruits list")
```

#output : Yes, 'apple' is in the fruits list

Python Lists

- Print the number of items in the list:

Example :

```
thislist = ["apple", "banana", "cherry"]
print(len(thislist))
#output : 3
```

Python Lists

- To add an item to the end of the list, use the append() method:
- Example :

```
thislist = ["apple", "banana", "cherry"]
thislist.append("orange")
print(thislist)
#output : ['apple', 'banana', 'cherry', 'orange']
```

Python Lists

- To add an item at the specified index, use the insert() method:
- Example :

```
thislist = ["apple", "banana", "cherry"]
thislist.insert(1, "orange")
print(thislist)
```

#output : ['apple', 'orange', 'banana', 'cherry']

Python Lists

- The remove() method removes the specified item:
- Example :

```
thislist = ["apple", "banana", "cherry"]
thislist.remove("banana")
print(thislist)
#output : ['apple', 'cherry']
```

Python Lists

- The pop() method removes the specified index, (or the last item if index is not specified):
- Example :

```
thislist = ["apple", "banana", "cherry"]
thislist.pop()
print(thislist)
#output : ['apple', 'banana']

thislist = ["apple", "banana", "cherry"]
thislist.pop(1) #remove let at 1
print(thislist)
#output : ['apple', "cherry"]
```

Python Lists

- The del keyword removes the specified index:

- Example :

```
thislist = ["apple", "banana", "cherry"]
del thislist[0]
print(thislist)
#output : ['banana', 'cherry']
```

Python Lists

- The del keyword can also delete the list completely:
- Example :

```
thislist = ["apple", "banana", "cherry"]
del thislist
print(thislist)
#output : Error
```

Python Lists

- The clear() method empties the list:
- Example :

```
thislist = ["apple", "banana", "cherry"]
thislist.clear()
print(thislist)
#output : []
```

Python Lists

- Make a copy of a list with the copy() method:
- Example :

```
thislist = ["apple", "banana", "cherry"]
mylist = thislist.copy()
print(mylist)
#output : ['apple', 'banana', 'cherry']
```

Python Lists

- Make a copy of a list with the list() method:
- Example :

```
thislist = ["apple", "banana", "cherry"]
mylist = list(thislist)
print(mylist)
#output : ['apple', 'banana', 'cherry']
```

Python Lists

- Join two list with +
- Example :

```
list1 = ["a", "b" , "c"]
list2 = [1, 2, 3]
list3 = list1 + list2
print(list3)
#output : ['a', 'b', 'c', 1, 2, 3]
```

Python Lists

- Append list2 into list1: (same as +)
- Example :

```
list1 = ["a", "b" , "c"]
list2 = [1, 2, 3]

for x in list2:
    list1.append(x)

print(list1)
#output : ['a', 'b', 'c', 1, 2, 3]
```

Python Lists

- Use the extend() method to add list2 at the end of list1: (same as append)
- Example :

```
list1 = ["a", "b" , "c"]
list2 = [1, 2, 3]
list1.extend(list2)
print(list1)
#output : ['a', 'b', 'c', 1, 2, 3]
```

Python Lists

- It is also possible to use the list() constructor to make a new list.
- Example :

```
thislist = list(("apple", "banana", "cherry")) # note the double
round-brackets
print(thislist)
```

#output : ['apple', 'banana', 'cherry']

Python Lists

- Sorting the list

Syntax : Syntax

***list*.sort(reverse=True|False,key=myFunc)**

reverse (Optional) : reverse=True will sort the list descending. Default is reverse=False

key (optional) : A function to specify the sorting criteria(s)

Python Lists

- Use sort() to sort the list
- Example :

```
mylist = ["banana", "cherry“, "apple"]
mylist.sort()
print(mylist)
#output : ['apple', 'banana', 'cherry']
```

Python Lists

- Use sort(reverse=True) to sort the list in reverse order.
- Example :

```
cars = ['Ford', 'BMW', 'Volvo']
cars.sort(reverse=True)
Print(cars)

#output : ['Volvo', 'Ford', 'BMW']
```

Python Lists

- You can Sort the list by the length of the values:

Example :

```
# A function that returns the length of the value:
def myFunc(e):
  return len(e)

cars = ['Ford', 'Mitsubishi', 'BMW', 'VW']

cars.sort(key=myFunc)
```

Python Lists

- Python has a set of built-in methods that you can use on lists.

Method	Description
append()	Adds an element at the end of the list
clear()	Removes all the elements from the list
copy()	Returns a copy of the list
count()	Returns the number of elements with the specified value
extend()	Add the elements of a list (or any iterable), to the end of the current list
index()	Returns the index of the first element with the specified value
insert()	Adds an element at the specified position

Python Lists

pop()	Removes the element at the specified position
remove()	Removes the item with the specified value
reverse()	Reverses the order of the list
sort()	Sorts the list